Comparative Study of Temporal Segmentation Methods for Radar-Based Kick Gesture Recognition

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18th Dec 2025

Motivation & Why Radar

Motivation

- Modern vehicles use **hands-free "kick-to-open" systems** for comfort and convenience.
- This is especially useful when the driver's hands are occupied with bags or luggage.
- The system must correctly detect a **real kick** and ignore **random leg or body movements**.

Why Radar Sensing

- Camera-based systems face **privacy issues** and do not work well in poor lighting.
- Infrared and ultrasonic sensors are **sensitive to weather and surface conditions**.
- **FMCW radar** works reliably in all lighting conditions and preserves user privacy.

Key Challenge

- A kick gesture happens only for a **short duration** inside a long radar signal.
- Therefore, **accurate temporal segmentation** is required before gesture classification.

Problem Statement & Project Objectives

Problem Statement

- Radar signals are **long in duration**, but the kick gesture occurs only briefly.
- Different people perform kicks **in different ways**, causing high variability.
- Normal movements near the vehicle can produce **false activations**.
- Without proper segmentation, the classifier may learn **irrelevant signal parts**.

Project Objectives

- Accurately locate the **kick gesture region** in long radar recordings.
- Study and compare **different temporal segmentation techniques**.
- Train a deep learning model for **robust kick gesture recognition**.
- Ensure **person-independent performance** using unseen users.

Radar Data & Signal Representation

Radar Data Description

- The system uses an **FMCW radar sensor** to capture motion information.
- The radar outputs **complex baseband signals** representing reflected waves from moving objects.
- Each recording contains information from **four signal channels**.

Signal Representation

$$X(t) = [I_{lower}(t),\ Q_{lower}(t),\ I_{upper}(t),\ Q_{upper}(t)]$$

- **I and Q components** represent the in-phase and quadrature parts of the signal.
- Lower and upper channels capture reflections from different antenna paths.
- Sampling frequency: **4000 Hz**
- Each recording length: **~12000 samples**

Labels

- **Valid Kick** → intentional kick gesture
- **Invalid Motion** → non-kick or background movement

Signal Preprocessing

- **Why Preprocessing Is Needed**
- Raw radar signals contain **noise**, **static reflections**, and **unwanted background components**.
- Different users produce signals with **different amplitudes**.
- Preprocessing ensures the data is **clean and consistent** before segmentation and classification.

Bandpass Filtering

$$f_{low} = 50\ \text{Hz}, \quad f_{high} = 1999\ \text{Hz}$$

- A bandpass filter is applied to keep only **motion-related frequencies**.
- Low-frequency clutter (ground, vehicle body) is removed.
- High-frequency noise is suppressed.

Amplitude Normalization

$$x_{norm}(t) = 2\frac{x(t) - x_{min}}{x_{max} - x_{min}} - 1$$

- Scales all signals to a common range **[−1, 1]**.
- Reduces variation caused by different kicking strengths.
- Helps the deep learning model learn **general patterns**.

Method 1 – Energy-Based Segmentation

Basic Idea

- A kick involves **strong and fast leg movement**.
- This movement produces a **high-energy response** in the radar signal.
- The goal is to find the time where the signal energy is maximum.

Magnitude Computation

$$m(t) = \sqrt{I_{lower}^2(t) + Q_{lower}^2(t) + I_{upper}^2(t) + Q_{upper}^2(t)}$$

- Combines all four radar channels into a single motion signal.
- Removes phase information and keeps **overall motion strength**

Short-Time Energy

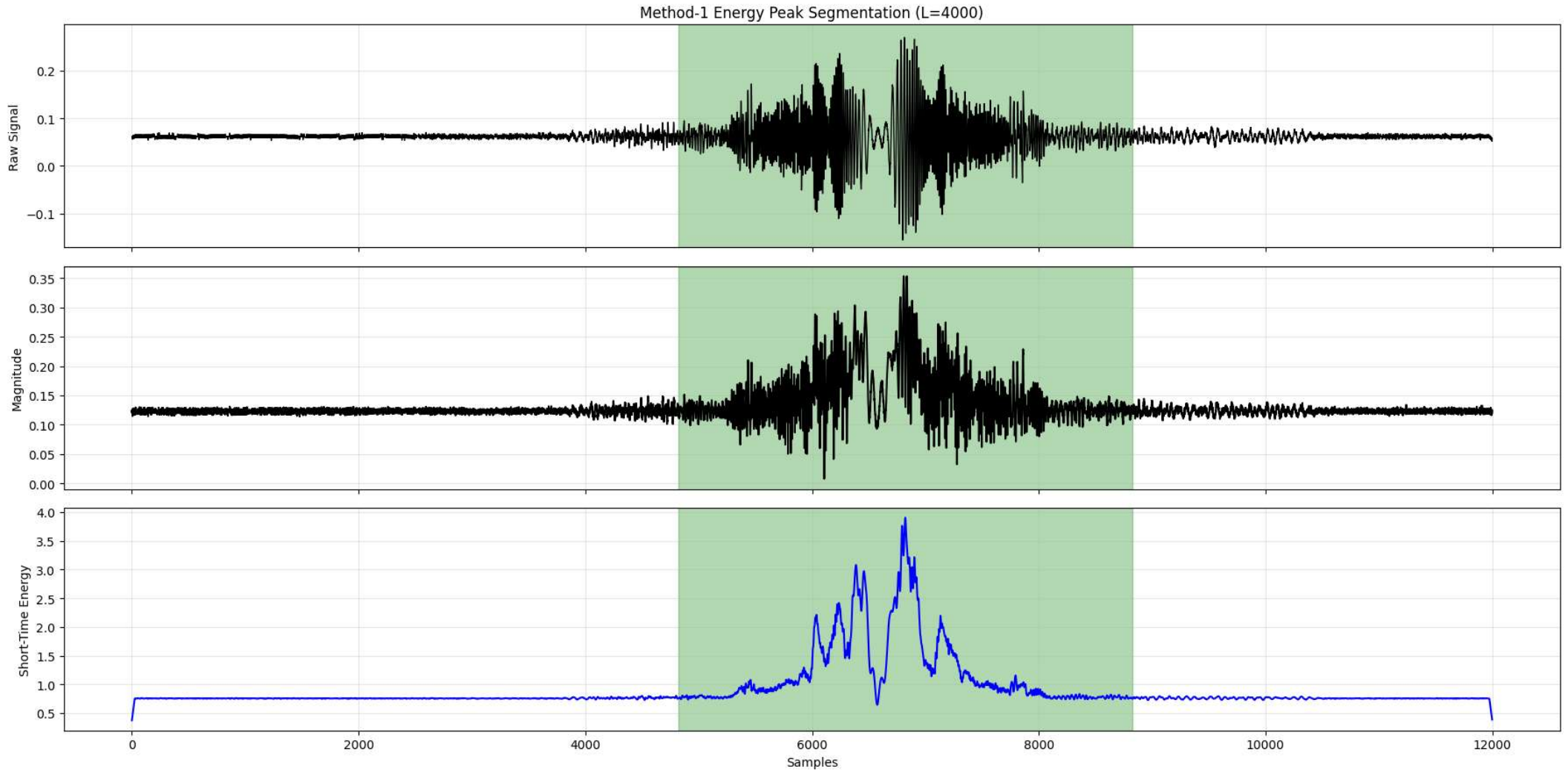
$$E(t) = \sum m^2(k)$$

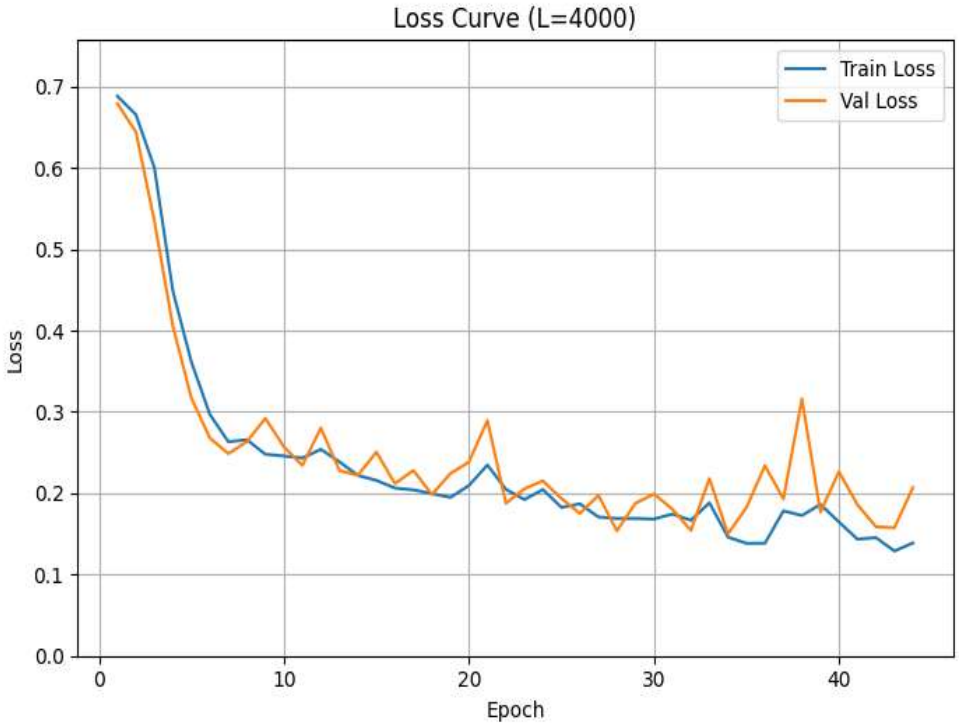
- Energy is computed over a short sliding window.
- Smooths noise and highlights **dominant motion regions**.

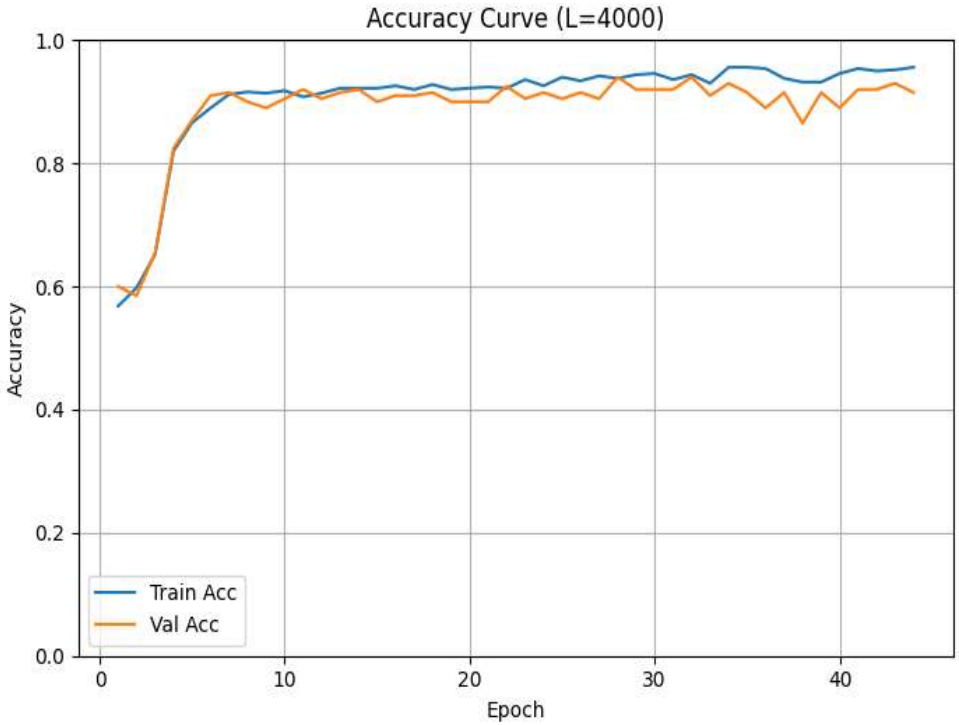
Segmentation Point

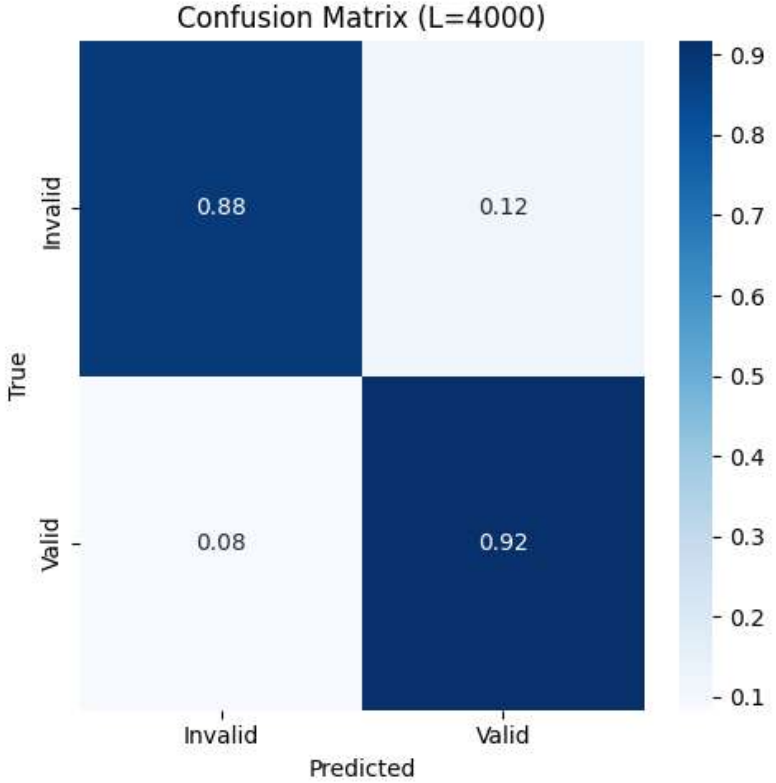
$$t^* = \arg\max E(t)$$

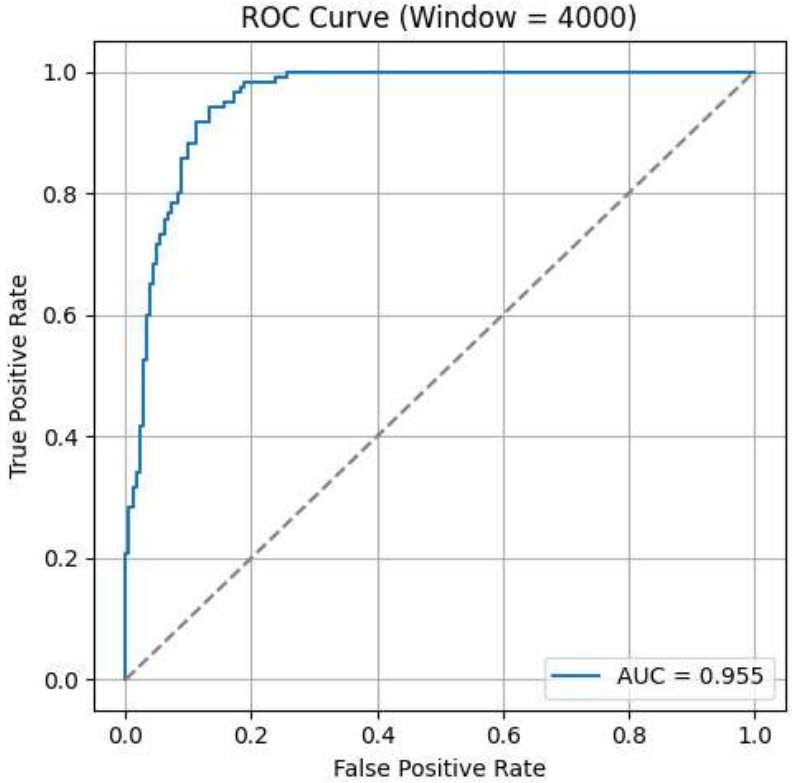
- The fixed-length window is extracted around t*.


Method-1 Energy Peak Segmentation (L=4000)
Raw Signal
Magnitude
Short-Time Energy
Samples
0
2000
4000
6000
8000
10000
12000


Loss Curve (L=4000)
Train Loss
Val Loss
Loss
Epoch
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0
10
20
30
40


Accuracy Curve (L=4000)
Train Acc
Val Acc
Accuracy
Epoch
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40


Confusion Matrix (L=4000)
Invalid
Valid
True
0.88
0.12
0.08
0.92
Invalid
Valid
Predicted
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1


ROC Curve (Window = 4000)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
AUC = 0.955

=== METHOD-1 FINAL TABLE [ENERGY PEAK SEGMENTATION] ===

	Window	ACC	Precision	Recall_TPR	FPR	AUC
0	4000	0.8967	0.8397	0.9167	0.1167	0.9548
1	2000	0.8933	0.8548	0.8833	0.1000	0.9432
2	1000	0.8867	0.8308	0.9000	0.1222	0.9283

Method 2 – Gradient-Based Segmentation

Basic Idea

- A kick does not only have high energy, it also **starts suddenly**.
- This sudden start creates a **sharp change** in the radar signal.
- Gradient-based segmentation focuses on **motion onset** rather than motion strength.

Temporal Gradient

$$g(t) = \left| \frac{dm(t)}{dt} \right|$$

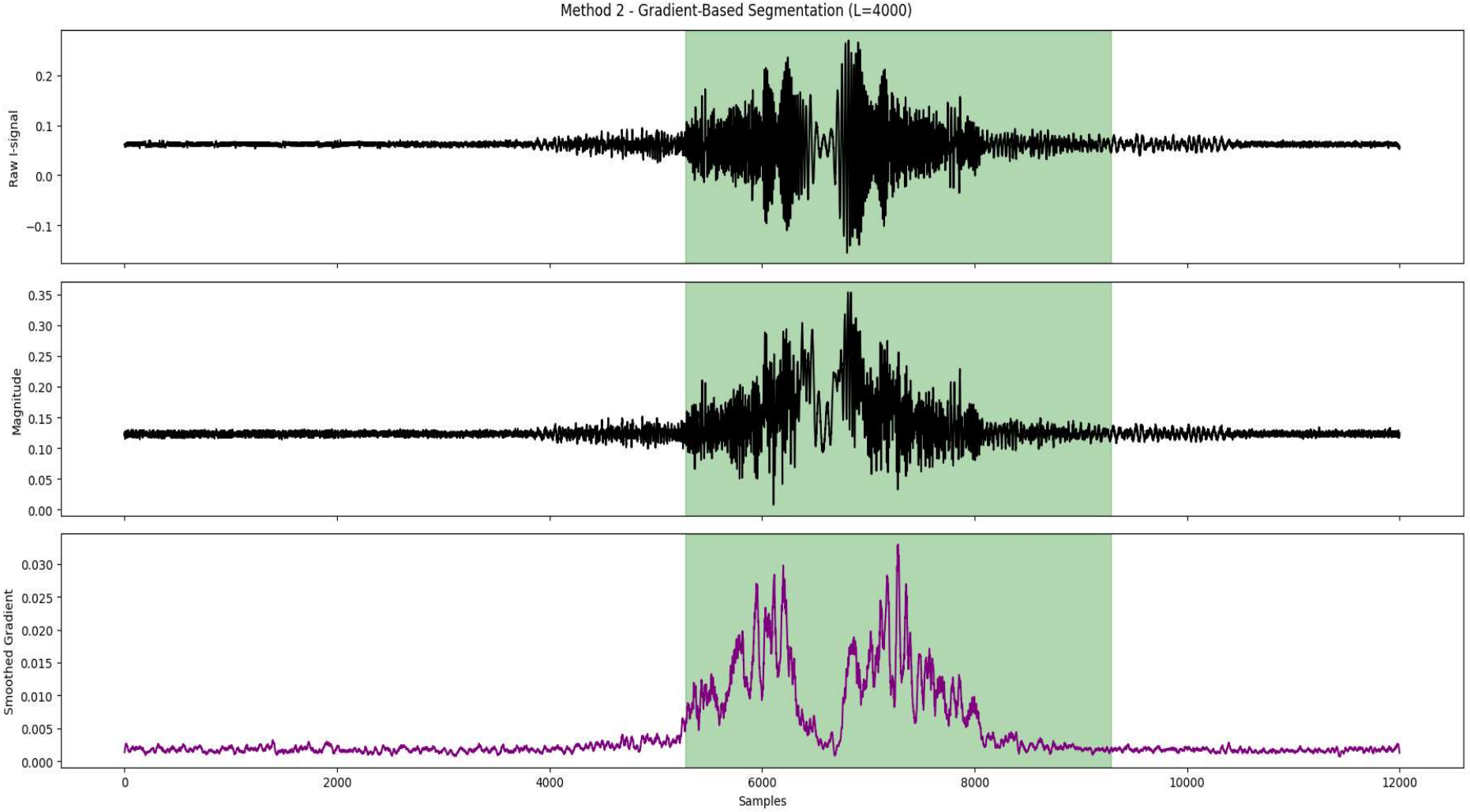
- Measures how fast the signal magnitude changes over time.
- Large gradient values indicate **sudden movement**.

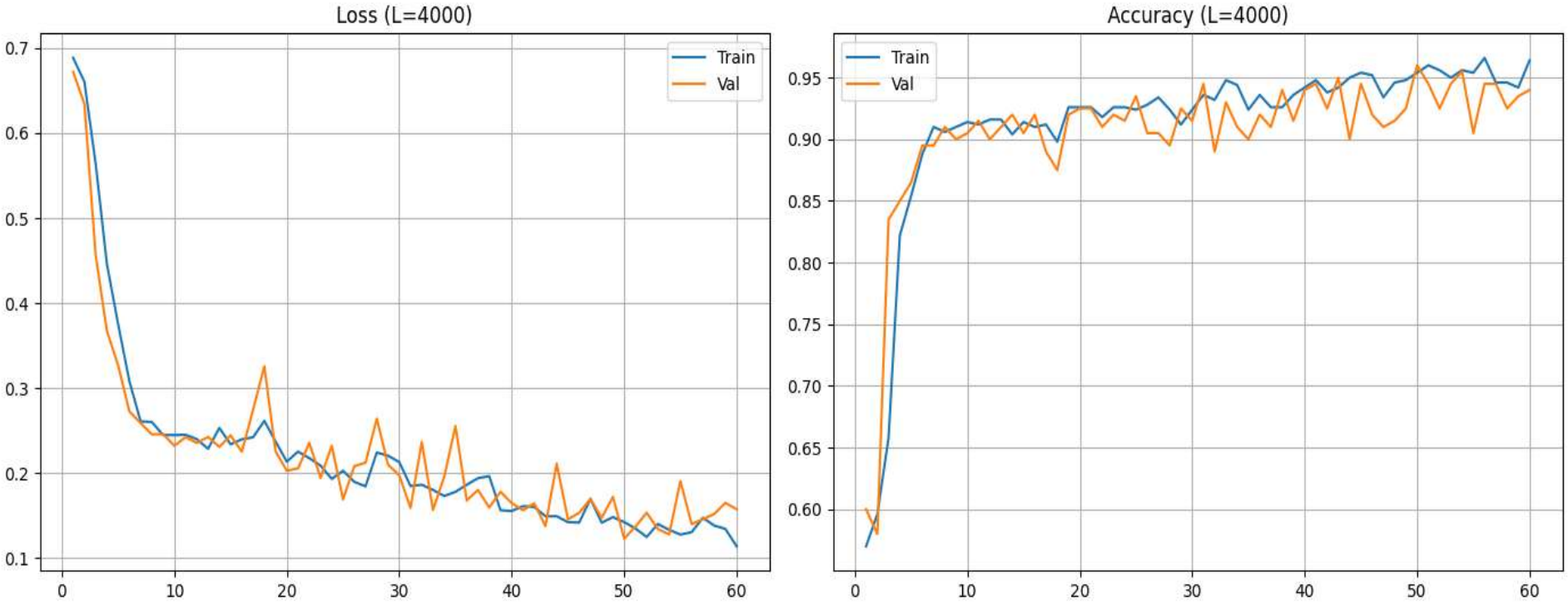
Gradient Smoothing

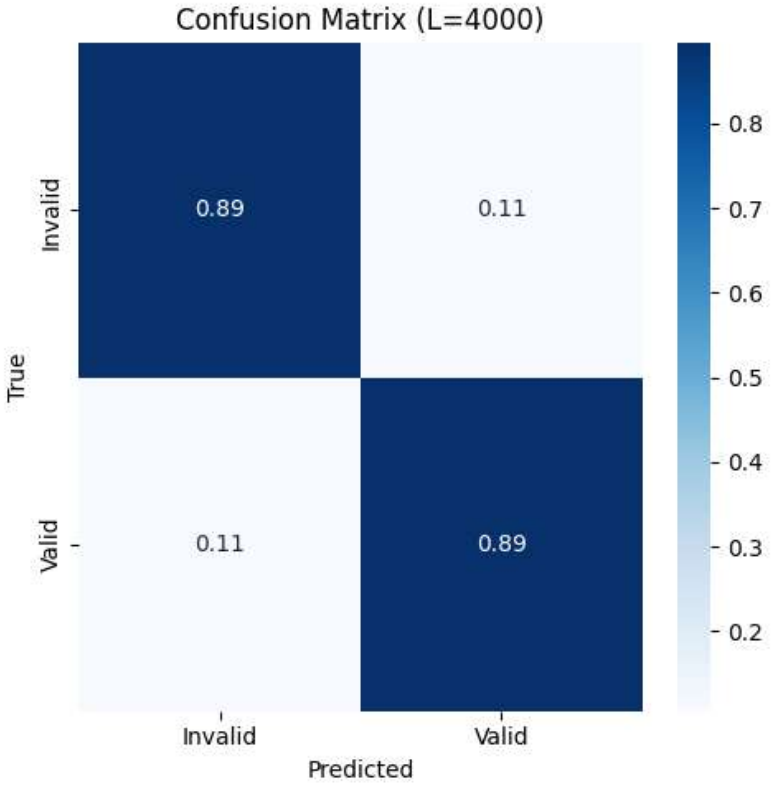
- A moving average is applied to reduce noise. $\tilde{g}(t) = \frac{1}{S}\sum g(k)$
- Ensures stable and reliable peak detection.

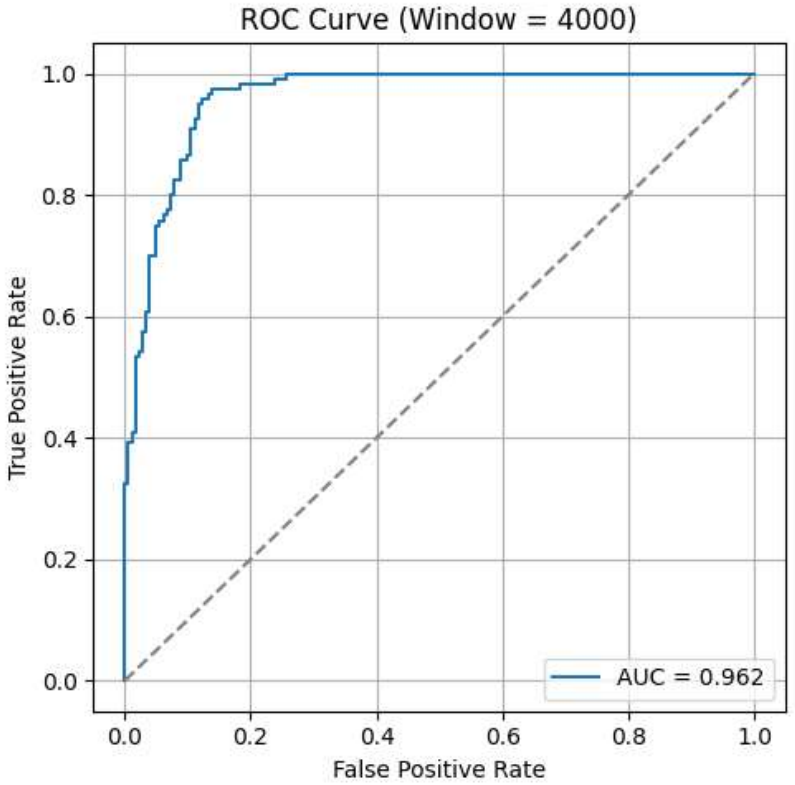
Segmentation Point $t^* = \arg\max \tilde{g}(t)$

- The fixed window is extracted around this point.


Method 2 - Gradient-Based Segmentation (L=4000)
Raw I-signal
Magnitude
Smoothed Gradient
Samples
0.2
0.1
0.0
−0.1
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.030
0.025
0.020
0.015
0.010
0.005
0.000
0
2000
4000
6000
8000
10000
12000


Loss (L=4000)
Train
Val
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
50
60
Accuracy (L=4000)
Train
Val
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0
10
20
30
40
50
60


Confusion Matrix (L=4000)
Invalid
Valid
True
0.89
0.11
0.11
0.89
Invalid
Valid
Predicted
0.8
0.7
0.6
0.5
0.4
0.3
0.2


ROC Curve (Window = 4000)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
AUC = 0.962

=== METHOD-2 FINAL TABLE [GRADIENT BASED METHOD] ===

	Window	ACC	Precision	Recall_TPR	FPR	AUC
0	4000	0.8933	0.8492	0.8917	0.1056	0.9619
1	2000	0.9033	0.8421	0.9333	0.1167	0.9451
2	1000	0.8933	0.8333	0.9167	0.1222	0.9356

•Shows how frequency (velocity) changes over time.

Method 3 – Doppler (STFT)-Based Segmentation

Basic Idea

- Radar measures **velocity** through the Doppler effect.
- A kick produces a **distinct velocity pattern** compared to normal movements.
- Doppler-based segmentation uses this **physical motion information**.

Short-Time Fourier Transform (STFT)

$$X(f,\tau) = \sum x(t)\, w(t-\tau)\, e^{-j2\pi ft}$$

•Converts the signal from time domain to **time–frequency domain**.
•Shows how frequency (velocity) changes over time

Doppler Energy

$$D(\tau) = \sum |X(f, \tau)|^2$$

- Measures motion energy in the Doppler domain.
- High Doppler energy indicates **strong foot movement**.

Segmentation Point

$$\tau^* = \arg\max D(\tau)$$

- The corresponding time location is used to extract the fixed window.

Windowing Strategy & Evaluation Approach

Fixed-Length Windowing

- After segmentation, a **fixed-length window** is extracted around the detected kick point.
- This window contains the most informative part of the kick gesture.
- Fixed-size input is required for training the deep learning model.

Window Lengths Tested

Window Length	Approx. Duration
4000 samples	~1.0 s
2000 samples	~0.5 s
1000 samples	~0.25 s

- Different window sizes capture different levels of motion context.

Training Strategy

- One segmented window is used per recording for training.
- Ensures clean and consistent learning.

Testing Strategy (Sliding Window)

- During testing, a **sliding window** is applied over the full signal.
- Predictions from multiple windows are aggregated for final decision.
- This follows the **real-world deployment scenario**.

CNN-Based Classification Model

Why Deep Learning

- Radar signals are **high-dimensional time-series data**.
- Handcrafted features may miss important motion patterns.
- A CNN can **automatically learn discriminative features** from raw signals.

CNN Architecture (Overview)

- **1D Convolutional layers** extract temporal motion patterns.
- **Pooling layers** reduce noise and dimensionality.
- **Global Average Pooling** summarizes learned features.
- **Fully connected layer** outputs the final prediction.

Model Output

- Binary classification:
  - **1 → Valid Kick**
  - **0 → Invalid Motion**

Loss Function

$$L = -\left[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})\right]$$

- Binary cross-entropy loss penalizes incorrect predictions.
- Encourages confident and correct classification

Results & Outputs

Performance Evaluation

- The performance of the system depends strongly on the **segmentation method** and **window length**.
- All models were evaluated using a **person-wise 80/20 split** with unseen subjects.

Key Outputs Generated

- Segmented radar signals for all three methods
- CNN training and validation curves
- ROC curves and confusion matrices
- Quantitative performance comparison tables

Key Observations

- **Energy-based segmentation**
  - Simple and fast
  - More false detections in background motion
- **Gradient-based segmentation**
  - Better temporal localization
  - More stable than energy-based method
- **Doppler-based segmentation**
  - Most robust and consistent
  - Best generalization across unseen users

Best Configuration

- **Best segmentation method:** Doppler-based (STFT)
- **Best window length:** 4000 samples

Conclusion & Future Work

Conclusion

- This project presented a **radar-based approach** for kick gesture recognition.
- Accurate **temporal segmentation** was shown to be critical for reliable performance.
- Among all methods, **Doppler-based segmentation** achieved the most robust results.
- The proposed system generalizes well to **unseen users**, validating real-world applicability.

Key Contributions

- Comparative analysis of **three temporal segmentation techniques**.
- Implementation of a **person-independent evaluation strategy**.
- Integration of segmentation with **CNN-based classification**.

Future Work

- Extend the system to **multi-class gesture recognition**.
- Collect data from **more users and environments**.
- Explore advanced deep learning models and sensor fusion techniques.

Thank You…